Mohammad Sobhan Kasaei

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Education

Sharif University of Technology — Oct 2021 – Sep 2026
Bachelor of Science in Industrial Engineering — Tehran, Iran

- CGPA: 19.03/20 (3.96/4.0)
- Last Year GPA: 19.75/20 (4.0/4.0)

Sharif University of Technology — Aug 2023 – Sep 2026
Minor in Economics — Tehran, Iran

- CGPA: 19.62/20 (4/4)

NODET (Sampad) — Jun 2018 – Jun 2021
Diploma in Physics & Mathematics — Tehran, Iran

- CGPA: 18.86/20

Publications

- **Kasaei, S.**, Radman, M., Asef-Vaziri, A., & Khalilzadeh, F. (2026). "Unidirectional Material-Handling Network Design with Empty Flows in Unequal-Area Facility Layout: Mathematical Formulation, Structural Analysis, and a Single-Solve Scalarization Method" *(In Preparation)*
- **Kasaei, S.**, Montazeri Hedesh, M. M., & Rohban, M. H. (2026). "Shared Representation Learning for Multi-Objective Neural Bandits with Random Scalarization" *(In Preparation)*

Experience

Quera Company — Jul 2024 – Feb 2025
Mentor — Artificial Intelligence Bootcamp — Tehran, Iran

- Mentored four teams of five students each through weekly sessions on core machine learning and deep learning concepts, providing guidance on assignments and projects.
- Designed and implemented a practical project using household income and expenditure survey data published by the Statistical Center of Iran, in which students formulated statistical questions, conducted causal analysis using econometric techniques, and performed predictive analysis using machine learning algorithms.

1&1 Company — Jul 2024 – Sep 2024
International Marketing Intern — Tehran, Iran

- Worked under the guidance of Prof. Majid Rafiee and Dr. Ghazimoradi at 1&1, one of Iran's largest and longest-established food production and agri-processing companies, whose products are exported to over 20 countries including the US, UK, Germany, Japan, and China.
- Assisted in identifying export and production opportunities in the agricultural sector, including potential partners and suppliers, market trend analysis, and guidance on regulations and logistics for international expansion.
- Collected and analyzed agricultural data by scraping datasets from FAO, EFSA, and various Ministries of Agriculture.

Sharif University of Technology Specialized Training Center — Jan 2023 – Jul 2023
Mentor — Data Science Introductory Course — Tehran, Iran

- Graded assignments and provided constructive feedback to students.
- Reviewed course material with students individually and in small groups.

Notable Course Projects

The Relationship Between Income Inequality and Crime Rates in U.S. States *2024*
Econometrics Course Project – Lecturer: Dr. Naser Amanzadeh

- Built a state-level panel of inequality and crime indicators from official U.S. sources.
- Replicated a Swedish econometric study using OLS/FEM models in STATA.
- Estimated the effect of inequality (Gini) on violent and non-violent crime with robustness checks.
- Interpreted results using standard economic theory on inequality and crime.

Analyzing the Causal Effect of Intercity Traffic on Voter Turnout *2024*
Regression Analysis Course Project – Lecturer: Dr. Kasra Alishahi

- Focused on Iran's 2016 parliamentary elections to investigate abnormal voter turnout across districts.
- Integrated intercity traffic counter data with electoral statistics to identify suspicious mobility patterns.
- Developed regression models to assess whether traffic volumes influenced turnout rates.
- Interpreted results in the context of political geography and voting behavior in Iran.

Simulation of Patient Flow in a Hospital *2024*
Fun of Comp Simul Course Project – Lecturer: Dr. Nafiseh Sedghi

- Developed a DES model of hospital flows including triage, labs, surgery, ICU/CCU, and wards.
- Implemented the simulation in Python with resource constraints and stochastic pathways.
- Calibrated arrivals and service times; validated behavior via tracing and stability checks.
- Evaluated key KPIs: queue lengths, waiting times, system time, and bed utilization.
- Ran sensitivity tests and compared capacity scenarios for stability improvements.

Prediction and Optimization of NYC Green Taxi Services *2024*
Transportation Planning Course Project – Lecturer: Dr. Erfan Hasannayebi

- Processed multi-year NYC taxi trip data and engineered spatial–temporal features.
- Built ML models to forecast zone-level hourly demand (RF, XGBoost).
- Formulated a MILP to optimize vehicle allocation and repositioning across zones.
- Solved using Pyomo–Gurobi and evaluated improvements over baseline strategies.
- Summarized insights and operational recommendations for fleet management.

Teaching Assistantships

Engineering Statistics — Head TA Spring 2024
Dr. Majid Khedmati & Dr. Zahra S. Hasanpour Jesri

- Designed and implemented, for the first time in the course, a project combining core course concepts with additional advanced topics; also taught these supplementary topics to students.
- Supervised a team of TAs, coordinated assignment, quiz, and exercise creation, and managed the grading and evaluation of all course components including projects.
- Organized and led weekly review and problem-solving sessions, oversaw content preparation for exercises, quizzes, and projects, and delegated responsibilities among TAs to ensure smooth course delivery.

Engineering Statistics — Head TA Fall 2024, Spring 2025
Dr. Zahra S. Hasanpour Jesri

- Continued supervising a team of TAs, coordinating assignments, quizzes, exercises, and project design and grading.

Fundamentals of Python Programming — Head TA Fall 2023
Eng. Alireza Kadivar

- Managed the overall course workflow, including consolidating and announcing final grades.
- Supervised TAs in grading assignments, quizzes, and exams, ensuring accurate and timely evaluation.
- Co-designed, oversaw, and coordinated project assignments; assisted in project evaluation and presentations alongside TAs.

General Economics Fall 2024
Dr. Mohammad Vesal

- Designed and graded all course assessments, including assignments, projects, quizzes, midterm, and final exam.
- Held weekly problem-solving and review sessions to support students' understanding of course material.
- Collaborated with the instructor to ensure alignment between assessments and learning objectives.

Other Courses — Teaching Assistant Fall 2022 – Sep 2026
Various Professors

- Data Mining Basics and its Applications, Microeconomics, Operations Research II, Statistical Quality Control, Advanced Programming, Engineering Economics.

Voluntary/Extracurricular Activities

GAMEIN Event, Sharif University Jun 2022 – Jun 2023
Data Analysis Team Member Tehran, Iran

- Cleaned and consolidated historical event data and maintained the data archive.
- Designed a dashboard for the closing event.
- During the event, defined and monitored key performance indicators (KPIs) using live data, and identified rule violations and bug exploitation among participants.

Rasta Scientific Association Jul 2023 – Aug 2023
Instructor / Member - Optimization Workshop Tehran, Iran

- Designed and delivered an LP workshop targeted at high-school students; taught LP fundamentals and posed exercises.

Selected Courses

Faculty of Ind. Eng.		**Faculty of Comp. Eng. / Sci.**		**Faculty of Econ. & Mgmt.**	
Operation Research I	18.3/20	Linear Algebra	19.4/20	General Economics	20/20
Operation Research II	19.5/20	Artificial Intelligence	20/20	Microeconomics	20/20
Decision Making	18.2/20	Machine Learning	20/20	Macroeconomics	20/20
Prod and Inv Contr	19.4/20	Regress Analysis	18.3/20	Econometrics	20/20
Eng Statistics	20/20	Stochastic Processes (Grad.)	20/20	Fund of Game Theory	18.7/20
Fun of Comp Simul	18.4/20	Deep RL (Grad.)	20/20	Financial Economics	18.7/20
Transportation Planning	20/20	Convex Optimization	19.7/20		

Technical Skills

- **Programming & Markup Languages:** Python, C++, MATLAB, R, SQL, HTML/CSS
- **Optimization & Statistics:** GAMS, Gurobi (Gurobipy), CPLEX, SCIP, Pyomo, OR-Tools, CVX, Stata, Minitab

Languages

Persian: Native
English: Full Professional Proficiency (TOEFL iBT: 109/120)
French: Elementary Proficiency
Arabic: Elementary Proficiency

References

Ardavan Asef-Vaziri
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Department of Systems and Operations Management
California State University, Northridge
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Kourosh Eshghi
Professor
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Sharif University of Technology
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Maryam Radman
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Sharif University of Technology
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Mohammad Vesal
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Graduate School of Management and Economics
Sharif University of Technology
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